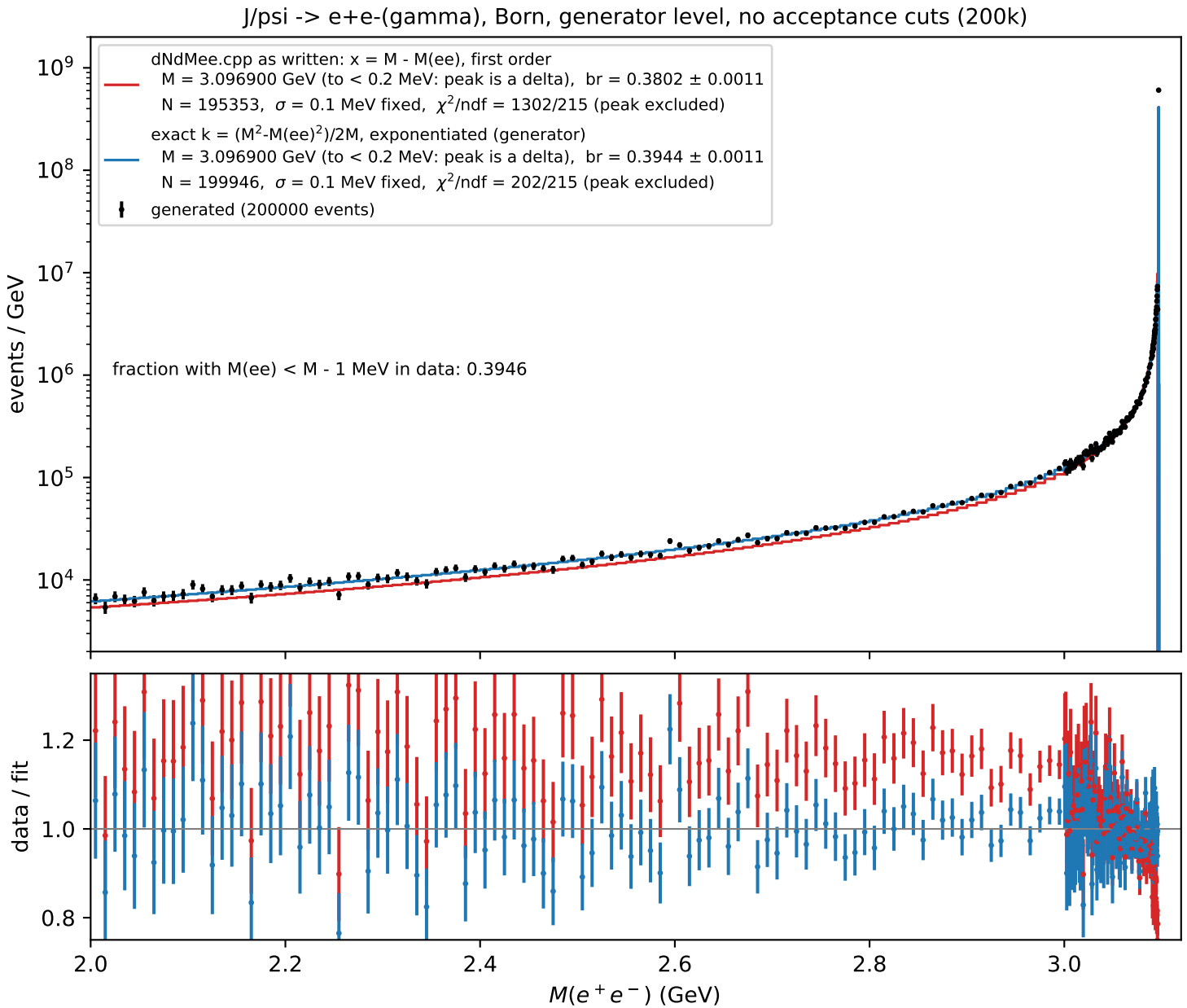
J/psi -> e+e-(gamma), Born, generator level, no acceptance cuts (200k)
dNdMee.cpp as written: x = M - M(ee), first order
M = 3.096900 GeV (to < 0.2 MeV: peak is a delta), br = 0.3802 ± 0.0011
N = 195353, σ = 0.1 MeV fixed, χ²/ndf = 1302/215 (peak excluded)
exact k = (M²-M(ee)²)/2M, exponentiated (generator)
M = 3.096900 GeV (to < 0.2 MeV: peak is a delta), br = 0.3944 ± 0.0011
N = 199946, σ = 0.1 MeV fixed, χ²/ndf = 202/215 (peak excluded)
generated (200000 events)
fraction with M(ee) < M - 1 MeV in data: 0.3946
events / GeV
data / fit
M(e⁺e⁻) (GeV)